roulette balls are sharing, her results and Gillian's results should match more than 25 percent of the time when they don't use corresponding roulette wheels:

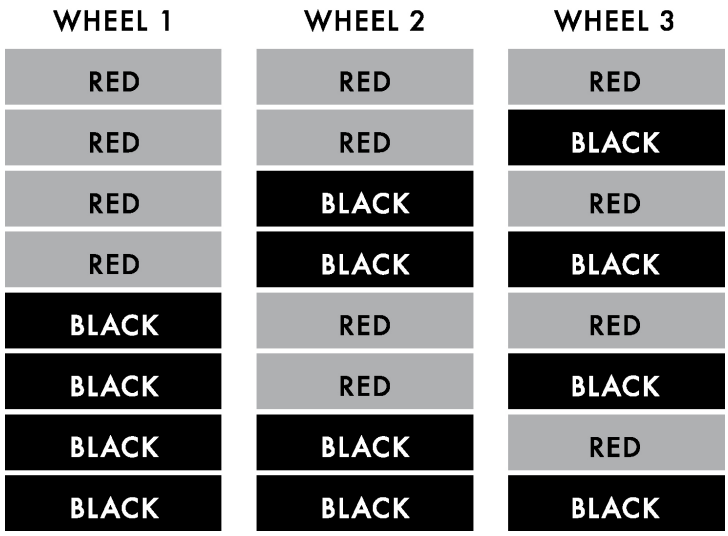


Figure 7.6. The only possible instruction sets for the roulette balls.

- If the two balls are sharing instruction sets Red Red Red or Black Black Black, then they'll match 100 percent of the time, even when they end up in differently numbered wheels.
- If the two balls are sharing one of the other instruction sets, then when Fatima and Gillian are using differently numbered wheels, the balls should land on the same color one-third (33 percent) of the time. For example, say the instruction set is Black Red Red. Then Fatima and Gillian will get different colors if they're using wheel combinations 1&2, 2&1, 1&3, or 3&1. But they'll get the same color if they're using wheel combinations 2&3 or 3&2—two out of the six total possibilities, or one-third. The other instruction sets (other than Black Black Black and Red Red Red) work the same way.

Therefore, when Fatima and Gillian aren't using corresponding roulette wheels, they should be getting the same color at least 33 percent of the time, because there are no instruction sets that would make such matches